Lab 1: Study of Different Types of Diode Characteristics

PN Junction Diode

1. What is a semiconductor diode?
A semiconductor diode is a two-terminal device formed by joining p-type and n-type semiconductor materials. It allows current to flow easily in one direction (forward bias) and blocks current in the opposite direction (reverse bias).

2. Why is a PN junction diode called a unilateral device?
Because it conducts mainly in only one direction. Its behavior is not the same for both polarities of applied voltage.

3. What is meant by depletion region?
It is the region around the PN junction where mobile charge carriers are absent due to recombination of electrons and holes. This region acts as a barrier to current flow.

4. What is barrier potential?
Barrier potential is the built-in potential difference across the depletion region that opposes further diffusion of majority carriers. For silicon it is about 0.7 V, and for germanium about 0.3 V.

5. What is forward bias in a diode?
Forward bias means connecting the p-side to the positive terminal and the n-side to the negative terminal of a supply. This reduces the barrier potential and allows current to flow.

6. What is reverse bias in a diode?
Reverse bias means connecting the p-side to the negative terminal and the n-side to the positive terminal. This increases the barrier potential and only a tiny current flows.

7. Why does current increase rapidly after cut-in voltage?
Once the applied forward voltage overcomes most of the barrier potential, a large number of majority carriers cross the junction. Then a small increase in voltage causes a large increase in current.

8. What is knee voltage (cut-in voltage)?
Knee voltage is the minimum forward voltage at which a diode starts conducting appreciably. For silicon it is approximately 0.7 V.

9. What is reverse saturation current?
It is the small reverse current that flows in a reverse-biased diode due to minority carriers. It is nearly constant until breakdown occurs.

10. Why is reverse current very small in a diode?
Because in reverse bias the depletion region widens and majority carriers are blocked. Only a small number of minority carriers contribute to current.

11. Which charge carriers are responsible for reverse saturation current?
Minority carriers.

12. What is meant by an ideal diode?
An ideal diode has zero resistance in forward bias and infinite resistance in reverse bias. It conducts perfectly in one direction and blocks perfectly in the other.

13. State the V-I characteristics of an ideal diode.
In forward bias, voltage across it is zero for any forward current. In reverse bias, current through it is zero for any reverse voltage.

14. Why does a silicon diode conduct after a certain forward voltage?
Because the applied voltage must first overcome the junction barrier potential. After that, majority carriers cross the junction easily and conduction starts.

15. What happens to junction resistance in forward bias?
It becomes very low.

16. What happens to junction resistance in reverse bias?
It becomes very high.

17. What is the static resistance of a diode?
Static or DC resistance is the ratio of voltage across the diode to current through it at a particular operating point:

$$R_{dc} = \frac{V}{I}$$

18. What is dynamic resistance of a diode?
Dynamic or AC resistance is the small-signal resistance at a particular point on the V-I curve:

$$r_d = \frac{\Delta V}{\Delta I}$$

19. Why are forward and reverse characteristics plotted on the same graph?
To compare the diode behavior in both biasing conditions and to observe conduction, blocking, knee voltage, and reverse saturation current in a single V-I characteristic.

20. Name some applications of PN junction diodes.
Rectifiers, clippers, clampers, switching circuits, demodulators, protection circuits, and logic circuits.

Zener Diode

21. What is a Zener diode?
A Zener diode is a specially designed heavily doped PN junction diode intended to operate in reverse breakdown region without damage.

22. How is a Zener diode different from an ordinary PN diode?
An ordinary diode is normally used in forward bias, while a Zener diode is designed to operate safely in reverse breakdown and maintain nearly constant voltage.

23. Why is a Zener diode heavily doped?
Heavy doping makes the depletion region thin and allows breakdown to occur at a lower, controlled reverse voltage.

24. What is breakdown voltage?
It is the reverse voltage at which a diode starts conducting heavily in reverse bias.

25. In which region does a Zener diode normally operate?
In the reverse breakdown region.

26. Why does Zener diode maintain nearly constant voltage in reverse breakdown?
Because once breakdown begins, a large change in current causes only a very small change in voltage across the diode.

27. Why is the dynamic resistance of a Zener diode low in the breakdown region?
Because in breakdown region the change in voltage is very small compared to the change in current:

$$r_z = \frac{\Delta V}{\Delta I}$$

So the slope resistance is low.

28. What is the main application of Zener diode?
Voltage regulation and voltage reference.

29. What is the difference between knee voltage and breakdown voltage?
Knee voltage refers to the forward voltage where normal diode conduction starts. Breakdown voltage refers to the reverse voltage where reverse conduction becomes large.

30. Can a Zener diode be used in forward bias? What happens then?
Yes. In forward bias it behaves like a normal silicon diode and conducts around 0.7 V.

31. Why is Zener diode used in voltage regulation?
Because it maintains an almost constant voltage across the load even when input voltage or load current changes, as long as it stays in breakdown region.

Lab 2: Study of Half Wave and Full Wave Diode Rectifier

Rectification Basics

1. What is rectification?
Rectification is the process of converting alternating current (AC) into direct current (DC).

2. Why is rectification needed?
Most electronic circuits require DC supply, but mains power is AC. So rectification is needed to obtain DC from AC.

3. What is the difference between AC and DC?
AC changes magnitude and direction periodically. DC flows in one direction and ideally has constant magnitude.

4. What is a rectifier?
A rectifier is an electronic circuit, usually using diodes, that converts AC into pulsating DC.

5. What are the main types of rectifiers?
Half-wave rectifier and full-wave rectifier. Full-wave rectifier can be center-tapped or bridge type.

Half-Wave Rectifier

6. What is a half-wave rectifier?
A half-wave rectifier is a rectifier that passes only one half-cycle of the AC input and blocks the other half-cycle.

7. How many diodes are used in a half-wave rectifier?
One diode.

8. Which half-cycle is conducted in a positive half-wave rectifier?
The positive half-cycle.

9. What happens during the negative half-cycle in a half-wave rectifier?
The diode becomes reverse biased, so ideally no current flows and output becomes zero.

10. Why is the output of a half-wave rectifier called pulsating DC?
Because the output is unidirectional but not constant; it rises and falls with time.

11. What is the average DC output of a half-wave rectifier?
For an ideal half-wave rectifier:

$$V_{DC} = 0.318V_m$$

where V_m is the peak value.

12. Why is a half-wave rectifier inefficient?
Because it uses only one half of the AC cycle and wastes the other half, so the output DC and efficiency are low.

13. What is the PIV of a diode?
PIV stands for Peak Inverse Voltage. It is the maximum reverse voltage a diode can withstand without breakdown.

14. What is the PIV requirement for a half-wave rectifier?
For a half-wave rectifier, the required PIV is equal to the peak input voltage.

15. What is the threshold voltage of a silicon diode?
Approximately 0.7 V.

Full-Wave Rectifier

16. What is a full-wave rectifier?
A full-wave rectifier converts both positive and negative half-cycles of AC into pulsating DC.

17. What are the types of full-wave rectifiers?
Center-tapped full-wave rectifier and bridge rectifier.

18. How many diodes are used in a bridge rectifier?
Four diodes.

19. How many diodes are used in a center-tapped full-wave rectifier?
Two diodes.

20. Why is a full-wave rectifier better than a half-wave rectifier?
Because it uses both half-cycles of input, giving higher average DC output, lower ripple, and better efficiency.

21. Why is the DC value of full-wave rectifiers higher than half-wave rectifiers?
Because both half-cycles contribute to the output, so the average value is larger.

22. Which diodes conduct in the positive half-cycle of the bridge rectifier?
According to your lab sheet, D2 and D3 conduct during positive half-cycle.

23. Which diodes conduct in the negative half-cycle of the bridge rectifier?
D1 and D4 conduct during negative half-cycle.

24. Why does load current remain in the same direction in the bridge rectifier?
Because the conducting diode pair changes every half-cycle in such a way that current through the load always flows in the same direction.

25. What is the ripple factor?
Ripple factor is the ratio of the RMS value of AC component in the rectifier output to the DC component:

$$r = \frac{V_{ac(rms)}}{V_{dc}}$$

It indicates how much ripple is present.

26. What is rectification efficiency?
It is the ratio of DC output power to AC input power:

$$\eta = \frac{P_{dc}}{P_{ac}}$$

27. What is the transformer utilization factor?
TUF is the ratio of DC power delivered to the load to the AC rating of the transformer secondary.

28. What are the advantages of bridge rectifiers?
No center-tapped transformer needed, better transformer utilization, full-wave rectification, and lower PIV per diode compared to center-tapped type.

29. What is a disadvantage of center-tapped full-wave rectifiers?
It requires a center-tapped transformer, which is bulkier and costlier.

30. Why is the output frequency of a full-wave rectifier twice the input frequency?
Because both half-cycles appear as positive pulses at the output. So for each input cycle, two output pulses are produced.

Filter

31. Why is a filter capacitor used in rectifier circuits?
To smooth the pulsating DC output and reduce ripple.

32. What happens when filter capacitance is increased?
Ripple decreases and the output becomes smoother and closer to pure DC.

33. Why is ripple reduced by a large capacitor?
A large capacitor stores more charge and discharges slowly through the load, so the output voltage falls less between peaks.

34. Compare output waveform for 1 µF and 2200 µF capacitor.
With 1 µF, smoothing is small and ripple remains noticeable. With 2200 µF, smoothing is much better and output becomes nearly steady DC.

35. Why can AC waveform and pulsating DC waveform not be observed simultaneously at different points with one oscilloscope channel?
Because one channel can measure only one point relative to ground at a time. To observe both simultaneously, separate channels or proper referencing are needed.

Lab 3: Study of BJT Amplifier (with Frequency Response) and Switch

BJT as Switch

1. What are the operating regions of a BJT?
Cutoff region, active region, and saturation region.

2. In which region does a BJT operate as an amplifier?
Active region.

3. In which regions does a BJT operate as a switch?
Cutoff and saturation regions.

4. What is the cutoff region?
The region where both junctions are reverse biased, base current is nearly zero, collector current is nearly zero, and transistor is OFF.

5. What is the saturation region?
The region where both junctions are forward biased, collector current is maximum for the circuit, and transistor is ON.

6. Why is the transistor OFF in the cutoff region?
Because there is insufficient base-emitter forward bias to allow base current, so collector current is essentially zero.

7. Why is the transistor ON in the saturation region?
Because enough base current is applied so that the transistor conducts heavily and behaves like a closed switch.

8. What happens to collector current in the cutoff state?
It becomes nearly zero.

9. What is the approximate value of V_{CE} in saturation?
Approximately 0.1 V to 0.3 V, commonly taken as 0.2 V.

10. Why is LED OFF in cutoff and ON in saturation?
In cutoff, no collector current flows through the LED. In saturation, collector current flows and lights the LED.

11. What is a DC load line?
It is a straight line on the I_C-V_{CE} graph showing all possible operating points for a given collector supply and collector resistance.

12. Why are only two operating points used for switch operation?
Because a switch needs only ON and OFF states, corresponding to saturation and cutoff.

13. What are the coordinates of the OFF point on the load line?
Approximately:

$$I_C \approx 0, \quad V_{CE} \approx V_{CC}$$

14. What are the coordinates of the ON point approximately?
Approximately:

$$V_{CE} \approx 0 \text{ or } 0.2V, \quad I_C \approx \frac{V_{CC}}{R_C}$$

or slightly less if $V_{CE(sat)}$ is included.

CE Amplifier

15. What is a common emitter amplifier?
It is a transistor amplifier in which the emitter terminal is common to both input and output circuits.

16. Why is CE amplifier widely used?
Because it provides high voltage gain, good current gain, and therefore good power gain.

17. Why is a CE amplifier called a voltage amplifier?
Because a small AC input voltage variation produces a much larger AC output voltage variation.

18. Why is output inverted in a CE amplifier?
When input increases, collector current increases, the drop across R_C increases, and collector voltage falls. So output varies opposite to input.

19. What is the phase difference between input and output in CE amplifiers?
180°.

20. What is the role of collector resistor R_C?
It converts collector current variations into output voltage variations.

21. What is the role of emitter resistor R_E?
It stabilizes the operating point against temperature and transistor parameter variations.

22. Why is an emitter resistor used?
For thermal stability and bias stabilization.

23. What is the purpose of a bypass capacitor?
It provides a low reactance path for AC around the emitter resistor, increasing AC gain while keeping DC stabilization.

24. What is the purpose of coupling capacitors?
They pass AC signals from one stage to another while blocking DC.

25. Why is proper biasing necessary in an amplifier circuit?
To place the transistor in active region and allow undistorted amplification.

26. What is Q-point or operating point?
The quiescent point is the DC operating condition of the transistor when no input signal is applied.

27. Why should Q-point be near the middle of the load line?
To allow maximum symmetrical swing of the output signal without clipping.

28. What causes clipping in CE amplifiers?
If the input is too large, the transistor reaches saturation in one half-cycle and cutoff in the other, causing distortion.

29. Why should the input signal be small in a CE amplifier?
So the transistor remains in active region and the output remains amplified without distortion.

30. What happens if the transistor enters saturation during amplification?
The output waveform gets clipped on one side.

31. What happens if the transistor enters a cutoff during amplification?
The output waveform gets clipped on the other side.

Frequency Response

32. What is the frequency response of an amplifier?
It is the variation of amplifier gain with frequency.

33. Why does gain fall at low frequencies?
Because coupling and bypass capacitors have high reactance at low frequency, reducing signal transfer and bypass action.

34. Why does gain fall at high frequencies?
Due to transistor internal capacitances and stray capacitances, which reduce gain.

35. What is the effect of coupling capacitors at low frequency?
Their reactance becomes high, causing voltage drop across them and reducing signal reaching the transistor.

36. What is the effect of stray capacitance at high frequency?
It shunts signal components and lowers gain.

37. What is meant by midband gain?
The nearly constant gain of the amplifier over the middle frequency range.

38. Why is semi-log graph used for frequency response?
Because frequency spans a very wide range. A logarithmic frequency axis makes the response easier to plot and interpret.

39. What is voltage gain?
Voltage gain is the ratio of output voltage amplitude to input voltage amplitude:

$$A_v = \frac{V_o}{V_i}$$

40. Why is gain often expressed in dB?
Because dB is convenient for large ranges, simplifies comparison, and is standard in frequency response analysis:

$$\text{Gain}(\text{dB}) = 20 \log A_v$$

Lab 4: Study of Operational Amplifier as Inverter, Inverting Amplifier, Adder, Integrator, Non-Inverting Amplifier, Voltage Follower

Basic Op-Amp

1. What is an operational amplifier?
An operational amplifier is a high-gain differential voltage amplifier with very high input impedance and low output impedance, used for amplification and mathematical operations.

2. Why is an op-amp called a linear device?
Because within its linear operating range, output is proportional to input.

3. Name some common op-amp ICs.
LM741, LM747, LM358, LM339, LM324, TL071, TL082, CA3140.

4. Why is LM741 popular?
Because it is simple, widely available, low-cost, and commonly used for learning basic op-amp circuits.

5. How many pins does LM741 have?
8 pins.

6. What is the function of pin 2 of LM741?
Inverting input terminal.

7. What is the function of pin 3?
Non-inverting input terminal.

8. What is the function of pin 4?
Negative supply terminal, $-V_{CC}$.

9. What is the function of pin 6?
Output terminal.

10. What is the function of pin 7?
Positive supply terminal, $+V_{CC}$.

11. What is the function of pins 1 and 5?
Offset null pins used to minimize offset voltage.

12. Why is pin 8 called NC?
Because it has no internal connection.

13. Why do op-amps often need dual supply?
To allow both positive and negative output swings around ground.

14. What happens if negative supply is not used?
The op-amp cannot produce significant negative output; operation becomes limited to single-supply range.

15. How do you identify pin 1 in an IC package?
By the notch or dot marking on the package. Pin numbering starts from pin 1 and proceeds counterclockwise.

Non-Inverting Amplifier

16. What is a non-inverting amplifier?
An op-amp circuit where the input is applied to the non-inverting terminal and the output is in phase with the input.

17. Why is the output of a non-inverting amplifier in phase with input?
Because the signal is applied to the non-inverting input, so the feedback arrangement amplifies it without phase reversal.

18. Write the voltage gain formula of a non-inverting amplifier.

$$A_v = 1 + \frac{R_f}{R_i}$$

19. What is the minimum gain of a non-inverting amplifier?
1.

20. If $R_f = R_i$, what is the gain?

$$A_v = 1 + \frac{R_f}{R_i} = 1 + 1 = 2$$

Inverting Amplifier

21. What is an inverter circuit?
An op-amp circuit that produces an output equal in magnitude but opposite in phase to the input, usually with gain -1.

22. Why is the output 180° out of phase in an inverter?
Because the input is applied to the inverting terminal, causing phase reversal.

23. Write the gain formula of an inverting amplifier.

$$A_v = -\frac{R_f}{R_i}$$

24. What is the gain when $R_f = R_i$?

$$A_v = -1$$

25. What happens when $R_f = R_i$?
The circuit acts as an inverter with unity magnitude gain; output is equal to input in magnitude but opposite in polarity.

26. Why is it called virtual ground at the inverting input?
Because due to negative feedback, the inverting input remains at nearly the same potential as the non-inverting input, often near 0 V, even though it is not physically connected to ground.

27. What is clipping in op-amp output?
Clipping is distortion where the output waveform is cut off at the positive or negative saturation limit.

28. Why does clipping occur when input amplitude is increased too much?
Because the amplified output tries to exceed the supply voltage limits, so the op-amp saturates.

Adder

29. What is an op-amp adder or summing amplifier?
It is an op-amp circuit that produces an output proportional to the algebraic sum of multiple input voltages.

30. Why is the adder output inverted in an inverting summing amplifier?
Because the inputs are applied to the inverting terminal through resistors.

31. Write the output equation of an inverting adder.
General form:

$$V_{out} = -R_f \left(\frac{V_1}{R_1} + \frac{V_2}{R_2} + \ldots \right)$$

For equal resistors:

$$V_{out} = -(V_1 + V_2 + \cdots)$$

32. What happens to output when one input is +5 V and another is sinusoidal?
The output becomes the inverted sum of the sinusoidal signal and +5 V. So the waveform shifts by a DC level and is inverted.

33. What happens if the second input is −5 V?
The output becomes the inverted sum including $-5V$, so the DC shift changes in the opposite direction.

Integrator

34. What is an op-amp integrator?
It is an op-amp circuit whose output is proportional to the time integral of the input signal.

35. What components are used in an integrator?
An input resistor and a feedback capacitor, with the non-inverting input usually grounded.

36. Why does a square-wave input produce a triangular output in an integrator?
Because integration of a constant positive or negative voltage gives a linearly increasing or decreasing ramp. A square wave alternates between constant levels, so the output becomes triangular.

37. Write the output equation of an integrator.

$$V_{out} = -\frac{1}{RC}\int V_{in}\,dt$$

38. Why is an integrator called a mathematical circuit?
Because it performs the mathematical operation of integration on the input signal.

Voltage Follower

39. What is a voltage follower or buffer?
It is a non-inverting op-amp circuit where the output is directly fed back to the inverting input, giving unity gain.

40. What is the gain of a voltage follower?
1.

41. Why is a voltage follower useful?
It isolates stages, prevents loading, provides high input impedance and low output impedance.

42. What is the phase relationship between input and output in a buffer?
They are in phase.

43. Why is the buffer used for impedance matching?
Because it draws very little current from the source due to high input impedance and can drive the load effectively due to low output impedance.